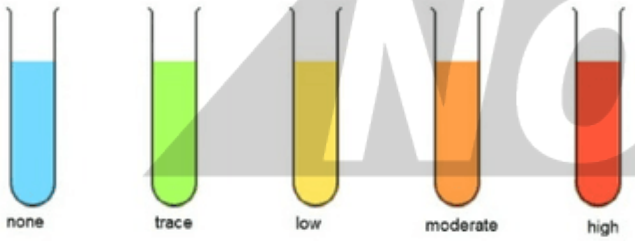
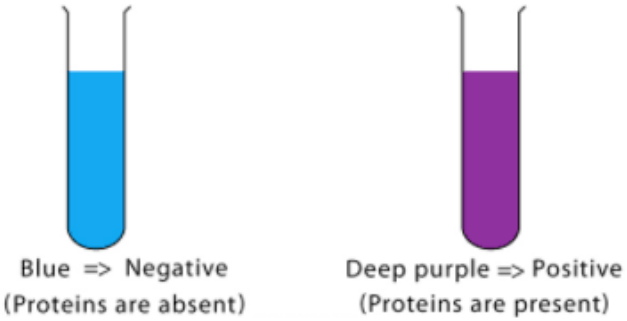
1. Biological Molecules

1.1. Testing for Biological Molecules

- Benedict's Test for Reducing Sugars:
  - An equal volume of sample and Benedict's solution is mixed
  - Heat in a water bath at a temperature above 80°C
    - **Positive:** green → yellow → orange → brick red
    - **Negative:** Benedict's reagent remains blue
- Benedict's Test for non-reducing sugars:
  - Hydrochloric acid is added to hydrolyse the sugar (if non-reducing sugar is present, it will break down into monosaccharides)
  - Heat in a water bath for approximately 2 minutes
  - Sodium hydroxide is added to make the solution alkaline.
  - Benedict's reagent is added, and Benedict's test is carried out
- Semi-quantitative Benedict's test:
  - Time taken for first colour change: measure the time taken for the colour to change from blue to green
    - The faster the colour change occurs, the greater the concentration of reducing sugar in the sample
  - Comparison to colour standards: Allow the Benedict's test to completely run through a sample with an unknown reducing sugar concentration
    - Compare the final colour of the solution with the colours obtained from carrying out Benedict's test on solutions with known reducing sugar concentrations to make an estimate.



- Biuret's test for proteins:
  - Equal amounts of the sample and Biuret's solution are added together (no heating required)
  - **Positive**: blue → purple
  - **Negative:** Biuret's reagent remains blue



- Emulsion test for lipids:
  - The sample is added to 2cm3 of ethanol and mixed well until it dissolves (lipids are soluble in ethanol)
  - This mixture is then placed into a test tube containing the same volume of cold water
  - **Positive:** milky emulsion
  - **Negative:** remains clear

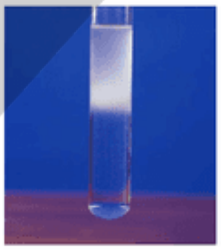


Fig 1. Test Tube Showing a Positive Emulsion Test for Lipids.

- Iodine test for Starch:
  - Add 2-3 drops of iodine to the liquid that is to be tested
    - **Positive:** orange-brown → blue-black
    - **Negative:** The iodine solution remains orange-brown

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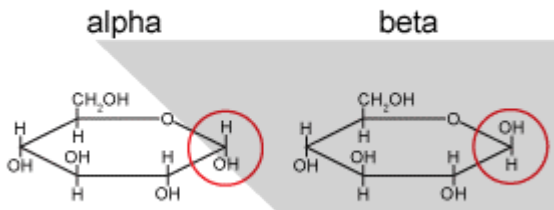
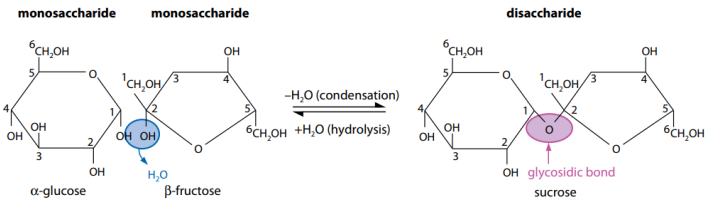
1.2. Carbohydrates and Lipids

- **Monomer**: a simple molecule that is used as a basic building block for the synthesis of a polymer; many monomers are joined together to make the polymer, usually by condensation reactions, e.g. monosaccharides, amino acids, nucleotides.
  - Glucose is a monomer (monosaccharide) with the molecular formula $C_6H_{12}O_6$
  - Two kinds of glucose, alpha-glucose and beta-glucose, depending on the position of the OH (hydroxyl) group in its ring structure



- **Polymer**: is a giant molecule made from repeating subunits of monomers, e.g. polysaccharides, proteins, nucleic acids
  - Lipids are **NOT polymers**, as they are not made of repeating subunits of monomers. They are made of 3 molecules of fatty acid and one molecule of glycerol, lacking a monomer unit
- **Macromolecules**: These are large and complex molecules that are formed due to the polymerisation of smaller monomers, e.g. polysaccharides, nucleic acids, lipids
- **Monosaccharide**: This molecule consists of a single sugar unit, the simplest form of carbohydrate and cannot be hydrolysed further. It has a general formula of $(CH_2O)n$
  - Examples: glucose (reducing sugar), fructose (reducing sugar)
- **Disaccharide**: a sugar molecule consisting of two monosaccharides joined together by a glycosidic bond.
  - Examples: sucrose (non-reducing sugar), maltose (reducing sugar)
- **Polysaccharide**: a polymer whose subunits are monosaccharides joined together by glycosidic bonds.
  - Examples: starch, glycogen, cellulose
- **Glycosidic bonds:** covalent bonds that occur between constituent monomers and are formed due to a condensation reaction involving removing a water molecule to form polysaccharides and disaccharides such as sucrose.
  - Hydrolysis (addition of water) can also separate these constituent molecules, which breaks the glycosidic bond between monomers.
  - Reverse of condensation
  - This is the reaction that occurs to break non-reducing sugars into reducing sugars when hydrochloric acid is added before Benedict's test



1.3. Polysaccharides

Starch:

- A macromolecule that is found in plant cells
- Polymer made up of glucose (monomer) subunits
- Contain 1,4 glycosidic bonds
- Highly compact and stores energy
- Made of two components: amylose and amylopectin

	Amylose	Amylopectin
Structure	α 1,4 glycosidic bonds	α 1,4 and α 1,6 glycosidic bonds, giving it its branched structure
Shape	Helical and more compact	Branched
Iodine test	Positive test with iodine	Negative test with iodine

Glycogen:

- A macromolecule that is used for the storage of energy in animal cells
- Polymer made from α glucose subunits
- The structure of glycogen is very similar to that of amylopectin; however, it contains more α 1,6 glycosidic bonds and is, hence, more branched

Cellulose:

- Found in the cell wall of plant cells
- Polymer made from β glucose units
- β-1,4 glycosidic bonds
- Alternate β-glucose molecules are rotated 180 degrees to form these bonds
- Hydrogen bonds are also formed between parallel cellulose molecules
- 60 and 70 cellulose molecules become tightly cross-linked to form bundles called microfibrils (10 nm).
- Microfibrils are, in turn, held together in bundles called fibres (50 nm) by hydrogen bonding
- Fibres increase tensile strength to withstand osmotic pressure, making the plant rigid and determining cell shape
- Freely permeable

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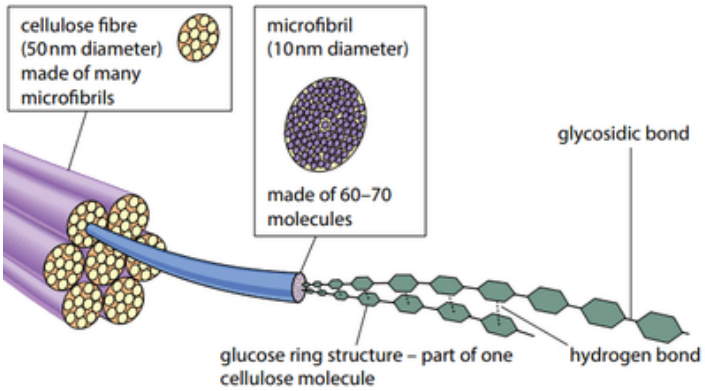
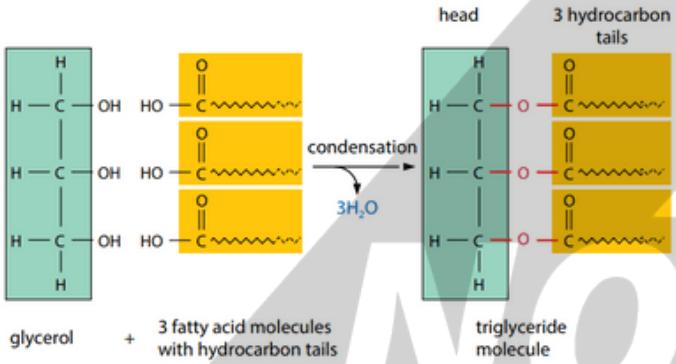


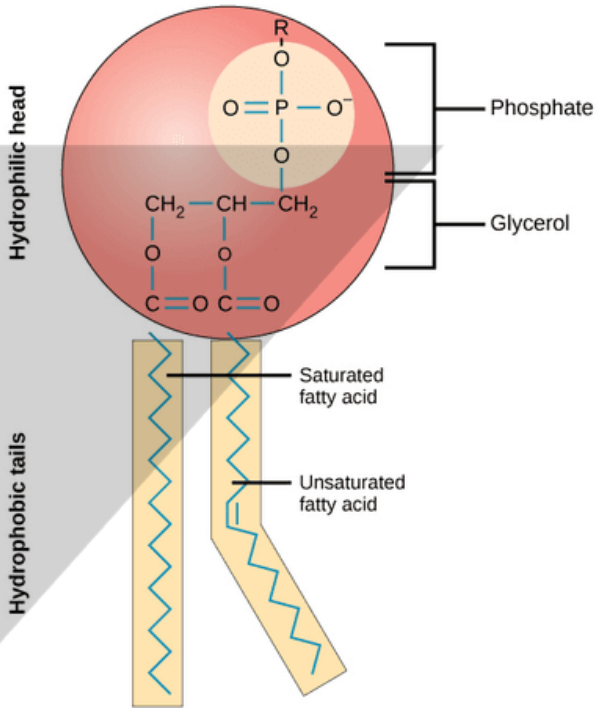
1.4. Triglycerides and Phospholipids

- Triglycerides:
  - Formed by condensing 3 fatty acid chains and a glycerol molecule
  - Joined by an ester bond
  - Fatty acid chains are long hydrocarbon chains with a carboxylic head; glycerol is an alcohol containing 3 OH groups.
  - Non-polar molecules
  - **Unsaturated fatty acids:** contain c=c bonds that are easier to break and melt easily. Vegetable oils.
  - **Saturated fatty acids:** contain c-c bonds that are solids at room temperature. Animal fats.



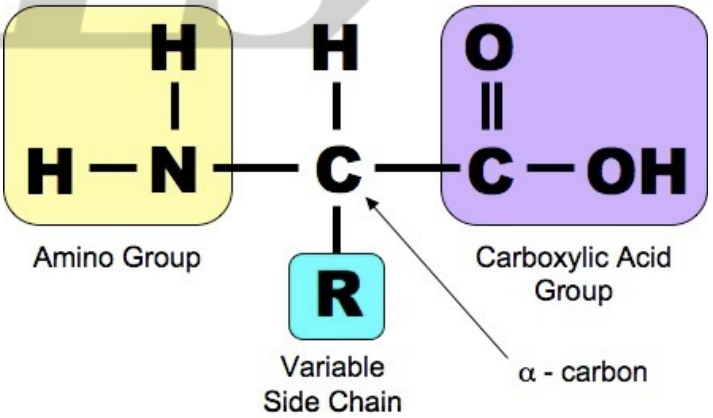
- Role of triglyceride:
  - Better energy reserves than carbohydrates as more CH bonds
  - Acts as an insulator and provides buoyancy
  - A metabolic source of water gives CO2 and H20 to oxidation in respiration

- Phospholipid:
  - The hydrophilic *head* contains a phosphate group and glycerol, while the hydrophobic *tail* contains 2 fatty acid chains.
  - The partial negative charge on the phosphate group gets attracted to the partial positive charge on the hydrogen atom of the water molecule and thus faces the aqueous environment.
  - Not an energy storage but component of cell membrane.



1.5. Proteins

Proteins: Made of amino acids which only differ in the R-groups/ variable side chains and will always contain an amine group (basic), a carboxyl group (acidic) and a hydrogen atom attached to the central carbon atom.



- A peptide bond is formed by condensation between 2 amino acids, forming a dipeptide.

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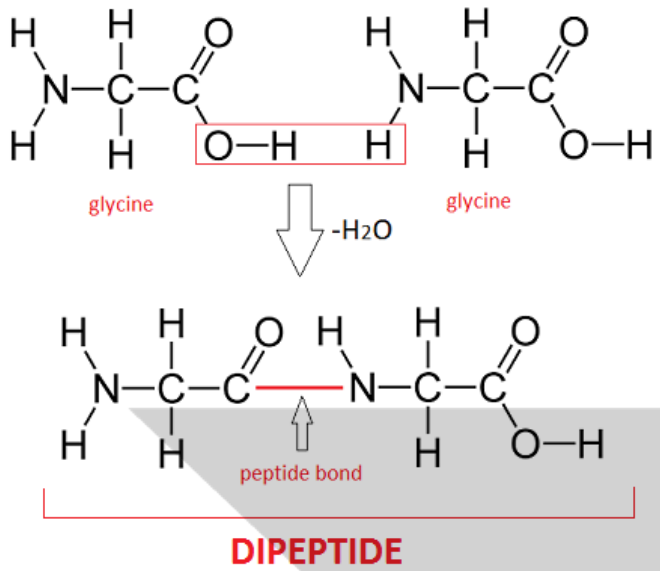
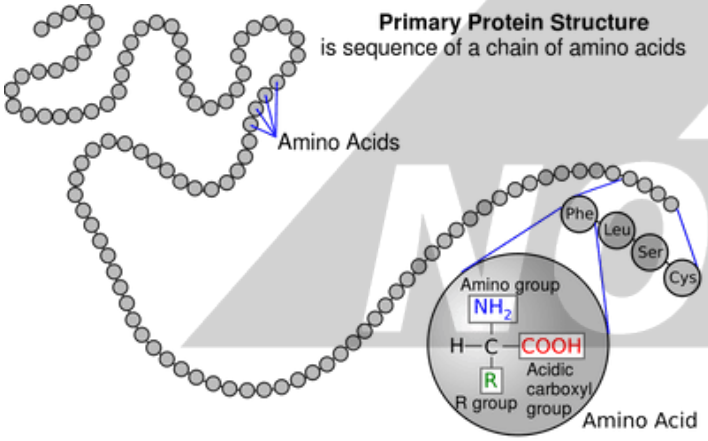
- Many amino acids that join together by peptide bonds form a polypeptide.



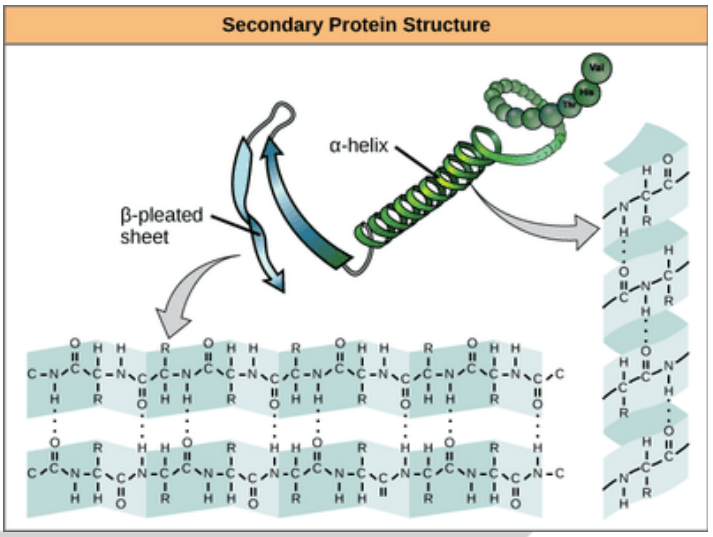
- Peptide bonds are broken when hydrolysed into amino acids, often occurring in the small intestine and stomach.

1.6. Protein Structure

- Primary structure:
  - The sequence of amino acids in a polypeptide/protein
  - A slight change in the sequence of amino acids can affect the protein's structure and function
  - It has a unique sequence for each protein.

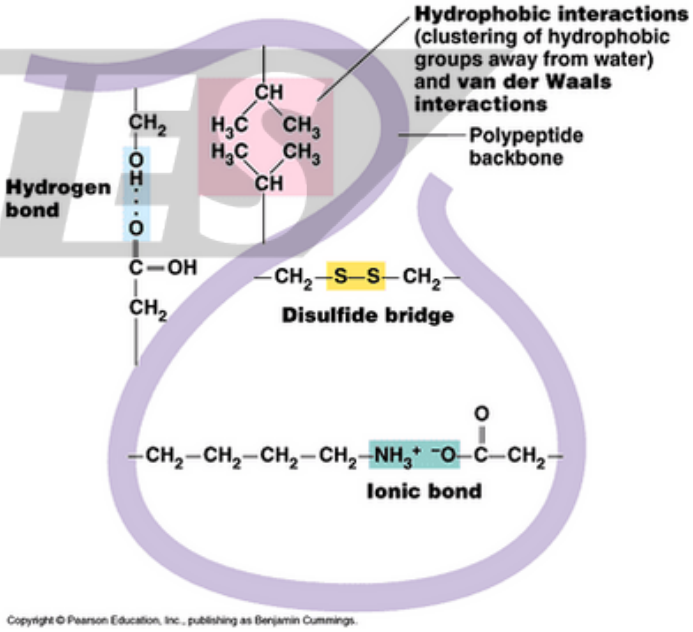


- Secondary structure:
  - The structure of a protein molecule resulting from the regular coiling or folding of the chain of amino acids
    - α- helix: the polypeptide chain twists into a regular spiral and is maintained by hydrogen bonds between the (-NH) group of one amino acid and the (CO-) group of another amino acid 4 spaces later in the polypeptide chain.
    - β- pleated sheet: the chain is not tightly coiled and lies in a looser, straighter shape.



- Tertiary structure:
  - The compact structure of a protein molecule results from the three-dimensional coiling of the already-folded chain of amino acids.
    - *Hydrogen bonds* between wide varieties of R-groups (can be broken by PH and temperature changes)
    - *Disulphide bridges* between two cysteine molecules (can be broken by reducing agents). Covalent bonds
    - *Ionic bonds* between R groups containing amine and carboxyl groups. (Can be broken by PH changes.)
    - *Hydrophobic interactions* between non-polar R groups.
    - In order of increasing strength:

      hydrophobic interactions < hydrogen bonds < ionic bonds < disulphide bridges



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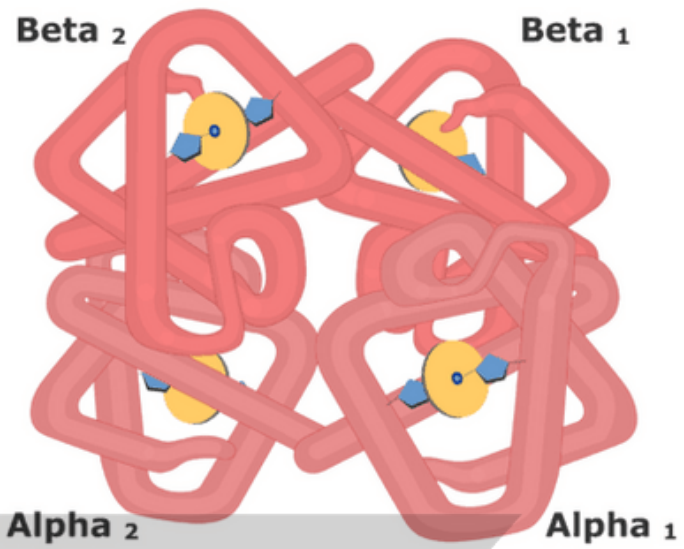
- Quaternary structure:
  - The three-dimensional arrangement of two or more polypeptides or a polypeptide and a non-protein component, such as haem, in a protein molecule
  - The polypeptide chains are held together by bonds in the tertiary structure.

1.7. Globular and Fibrous Proteins

- Globular proteins:
  - Curl up into a spherical shape with their hydrophobic regions pointing into the centre of the molecule and hydrophilic regions pointing outwards
  - They are soluble in water, e.g. enzymes and haemoglobin.
- Fibrous proteins:
  - Form long strands, are insoluble in water, and have structural roles, e.g. collagen, hair, and nails.

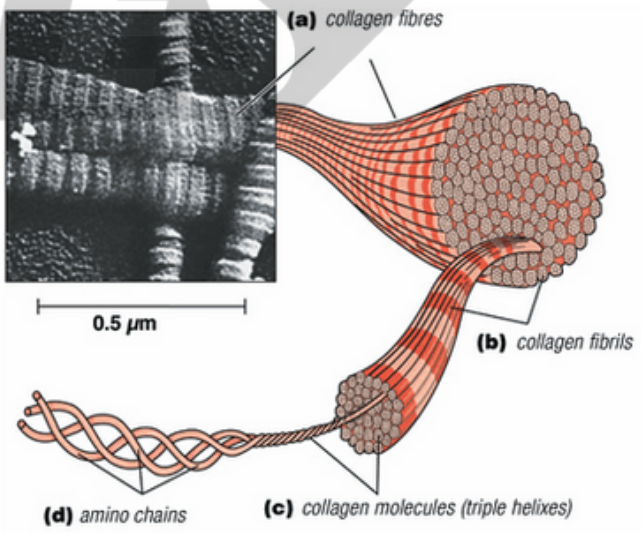
Haemoglobin

- A globular protein with a quaternary structure
- made up of 4 polypeptide chains, each with a globular tertiary structure
  - 2 beta globin & 2 alpha globin chains
- Each chain has a prosthetic (haem) group containing iron.
  - Each haemoglobin with the four haem groups can therefore carry four oxygen molecules (eight oxygen atoms)
- Function
  - Haemoglobin is responsible for binding oxygen in the lungs and transporting the oxygen to tissue to be used in aerobic respiration
  - Oxyhaemoglobin is formed when haemoglobin is bonded to oxygen
    - It is bright red when the haem group is combined with oxygen; otherwise, it's purplish.
- *Structure in relation to function*
  - The hydrophilic group allows haemoglobin to form hydrogen bonds with water
  - Each polypeptide chain has one haem group, meaning each can bond to an oxygen molecule.
  - Its shape allows it to pick up oxygen when the concentration is high, and to release it when the concentration is low



Collagen

- Collagen is the most common structural protein found in vertebrates
- It is a fibrous protein; insoluble.
- Collagen molecules are made up of three polypeptide chains that are wound (twisted) together, held by hydrogen bonds to form a triple helical shape with regular coils.
- Collagen fibrils are formed from collagen molecules, lying parallel to one another, held together by hydrogen bonds and covalent cross-links.
- Collagen fibres are made of many collagen fibrils grouped together, reinforced by additional covalent cross-links.
- Function
  - Collagen is a flexible fibrous protein that forms connective tissues
  - It is insoluble and has a high tensile strength, and can be stretched without breaking



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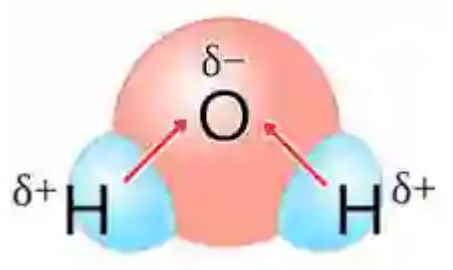
1.8. Water

- Water is a polar molecule, due to the unevenly distributed charge
- A water molecule contains two hydrogen atoms and one oxygen atom held together by hydrogen bonds



Properties of Water

- Solvent
  - Polar/charged molecules (e.g. sugars, glycerol) dissolve in water because it is polar.
    - Non-polar molecules (hydrophobic, e.g. oil) do not dissolve.
  - Hydrogen bonds form between water and solutes: H^+ dipoles attract negative ions, O^- dipoles attract positive ions.
  - Water is therefore an effective solvent, transport medium, and reagent (main component of cytosol).
  - Water is also a good medium for chemical reactions
- High Specific Heat Capacity
  - Water has high SHC due to the energy required to break its hydrogen bonds
    - A large amount of energy is needed to raise 1 kg of water by 1 °C.
  - Keeps organism temperature stable and slows temperature changes in water bodies, creating stable habitats.
    - Prevents enzyme denaturation from rapid temperature fluctuations.
- High Latent Heat of Evaporation
  - Water has a high latent heat of vaporisation because hydrogen bonds need to be broken before water can be vaporised.
    - So a lot of energy is taken up when breaking the hydrogen bonds
  - This allows a large amount of energy to be lost for a small amount of water, allowing sweating to be used as a cooling mechanism.
  - Also prevents bodies of water from being easily dried up.
- Density and Freezing Points
  - Ice is less dense than water and floats on it, insulating the water and preventing it from freezing, preserving aquatic life underneath it
  - Changes in the density of water with temperature cause currents, which help to maintain the circulation of nutrients in the oceans.

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